

WASP-43b

R-2642

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140105 · created 2026-08-29T01:30:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456662.85252 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.057
Transit depth (Rp/Rs)^2	3.91 +/- 2.24 [%]
Orbital Inclination (inc)	86.87 +/- 2.9
Airmass coefficient 1 (a1)	0.96 +/- 0.11
Airmass coefficient 2 (a2)	0.018 +/- 0.04
Scatter in the residuals of the lightcurve fit is	10.96 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.0
Transit Duration (day)	0.0614 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.057 vs 0.1594 (deviation 0.68σ)
Duration fit vs published	0.0614 d vs 0.0512 d (deviation 2.03σ)
Mid-time O–C	1.4 min
Depth SNR	3.5
Residual scatter	10.96 %

Light curve

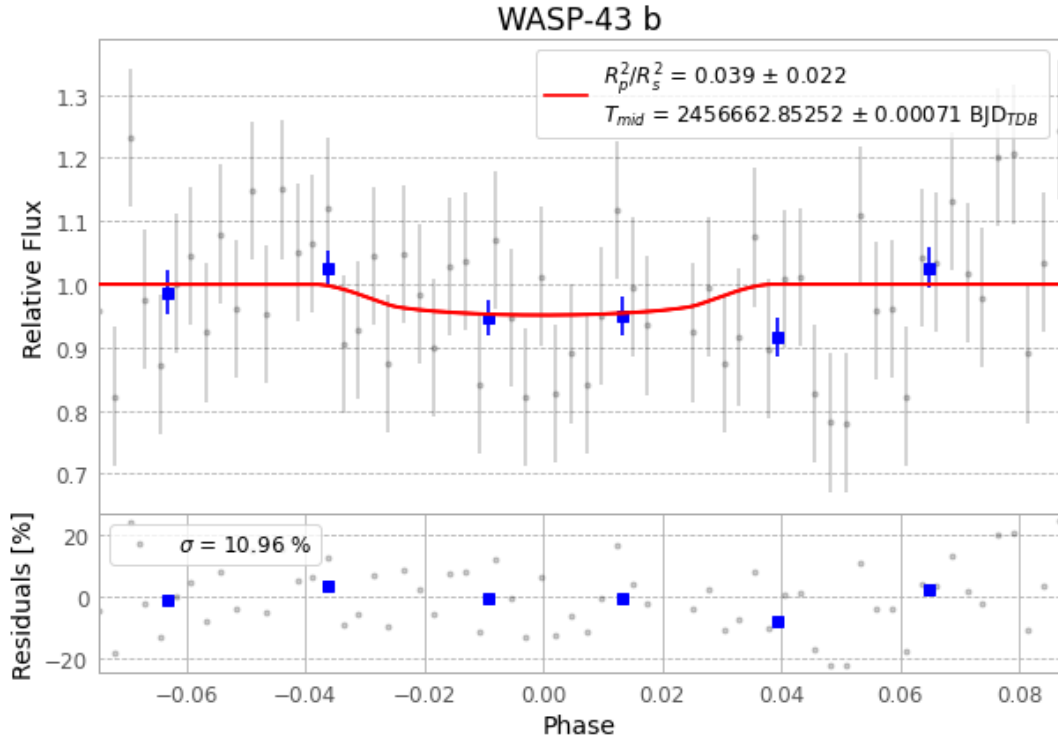


Figure 1 — FinalLightCurve_WASP-43 b_2014-01-04.png (SHA-256 9cf26ebfda48da70...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [557, 325], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2ae66432e9a03d5f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140105065412.FITS → WASP-43140105100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140105.log (e03a94621e17...), FinalLightCurve_WASP-43 b_2014-01-04.png (9cf26ebfda48...), FinalLightCurve_WASP-43 b_2014-01-04.csv (6aa4b26893a4...)

Compute resources

Wall time 164.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 01:30Z.